

Assignment 1: Pizza Pi – Iteration

Assignment for COSC 1100 – Intro to Programming

Prior to attempting this problem, you should have done the following:

1. View the Content, including assigned readings for weeks 1 through 6.
2. Viewed the Control Structures - Iteration videos from week 6 and completed the associated Pre-Class Activity.

General Requirements

1. This assignment is to be completed **individually**. Group submissions will not be considered.
2. Analyze the problem, design a documented plan (i.e. flowchart or pseudo-code), code and test a solution **following the step-by-step approach** presented in this course.
3. Submit your solution and plan to the appropriate assignment folder on DC Connect **by the due date provided**.
4. Your instructor will assign a grade and post feedback on your submission to DC Connect.

Program Requirements

The application should prompt the user to enter a diameter of a pizza in inches and then present the area of each pizza slice based in multiple possible configurations. The user should then able to either enter another diameter or indicate that they want to quit by entering '0'. Pizzas may be cut into up to 6, 8, 10, 12, or 16 slices dependent on the pizza size. The application should prompt for entry of a diameter and only process and display results if the entered diameter is in the range of 8" to 24" inclusive. When an invalid diameter is entered, (other than 0 to end) a message should be displayed and the user should be re-prompted for entry of a valid diameter.

Sample Input and Output:

```
Please enter the diameter of your pizza (0 to end program): 23.5
Pizza Diameter: 23.5"
Cut in 6 slices results in an area of 72.29" per slice.
Cut in 8 slices results in an area of 54.22" per slice.
Cut in 10 slices results in an area of 43.37" per slice.
Cut in 12 slices results in an area of 36.14" per slice. Cut
in 16 slices results in an area of 27.11" per slice.

Please enter the diameter of your pizza (0 to end program): 12
Pizza Diameter: 12"
Cut in 6 slices results in an area of 18.85" per slice. Cut
in 8 slices results in an area of 14.14" per slice.

Please enter the diameter of your pizza (0 to end program): 4
ENTRY ERROR
Pizza must have a diameter in the range of 12" to 36" inclusive.
Please try again.

Please enter the diameter of your pizza (0 to end program): 17.25
Pizza Diameter: 17.25"
Cut in 6 slices results in an area of 38.95" per slice.
Cut in 8 slices results in an area of 29.21" per slice.
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Cut in 10 slices results in an area of 23.37" per slice.

Cut in 12 slices results in an area of 19.48" per slice.

Please enter the diameter of your pizza (0 to end program): 0

Detailed Requirements

1. Prompt the user to enter the diameter of a pizza in inches. Pizza diameters can be entered as real numbers. Ensure that the user input is numeric using "try ... except" or other means.
2. When the user input is numeric, use a compound condition test to check that the diameter entered is within the range of 8" to 24" inclusive. If the diameter is not numeric, display an error message and re-prompt for a valid entry.
3. When the diameter entered is not numeric (i.e. cannot be "parsed" into a usable numeric value), display a message indicating the nature of the problem. When the diameter entered is numeric but outside of the allowable range, display an error message indicating that the input must be between 8" and 24" and reprompt for a valid entry. Do not allow any processing to continue with an entry of a diameter outside of the allowable range.
4. When the diameter entered is good, use a loop to determine and display all possible numbers of slices that this size of pizza can be sliced into, along with the area of each slice for each of these. This will be based on the following:
 - Diameters of 8" to < 12" can only be cut into six slices.
 - Diameters of 12" to < 14" can be cut into 6 or 8 slices.
 - Diameters of 14" to < 16" can be cut into 6, 8 or 10 slices.
 - Diameters of 16" to < 20" can be cut into 6, 8, 10, or 12 slices.
 - Diameters of 20" to 24" can be cut into 6, 8, 10, 12 or 16 slices.

This number of slices must be stored and displayed as a whole number only.

5. For each of these slice configurations, the output should include the number of slices, as a whole number, to be cut from that pizza as well as the area of each slice as a real number rounded to two decimal places. The format should match the example provided as closely as possible.
6. Continue to prompt for entry of pizza diameters, also prompting the user to enter '0' if they want to end the program. For each valid entry that isn't 0, display results for each entered diameter. When the user does enter '0' for diameter, immediately end the program without any further processing, message or prompt.

Style Guide Notes

To be eligible for full marks on this or any assignment in this course your application must conform to the requirements as outlined above as well as our prescribed style guide, in this case making sure to observe the PEP8 naming conventions for Python as well as appropriate and complete program documentation.

Development Hints

You may be tempted to utilize parts of your existing work or demo code as a starting point for this project. Be careful because while some things may be relevant, it is a different problem and there will be big differences! It is strongly recommended that you code this assignment following the "from plan to program" process from the beginning.

Be advised that an ideal solution to this problem may involve the declaration/assignment of multiple constants; possibly 10 or more.